

# ECE 447

## Lesson 36

### Binary System

### Performance, Part 1



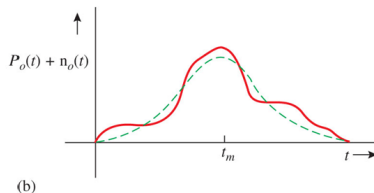
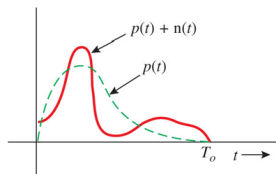
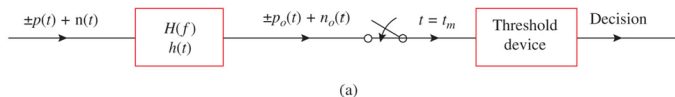
UNITED STATES  
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ACADEMY

# SCHEDULE AND ADMIN

- **Schedule.**
  - Lesson 36 - Matched filters, binary digital system performance (Part 1)
  - Lesson 37 - Binary digital system performance (Part 2)
  - Lesson 38 - Error correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
  - **Final Exam** – 0730-1120, Saturday, 12 Dec
- **Admin**
  - **HW9.** Assigned today. Due Lesson 40 (Mon, 7 Dec) to Gradescope.
  - **Lab 7.** Due Lesson 37 (Mon, 23 Nov) to Gradescope.

# BINARY COMMUNICATION SYSTEMS

- Received signal is  $y(t) = \pm p(t) + n(t), 0 \leq t \leq T_b$
- Receiver filters signal through  $H(f)$ :  $r(t) = h(t) * y(t) = \pm p_o(t) + n_o(t)$   
(ideally the filter aligns  $r(t)$  so it is sampled at the "widest opening of the eye")
- Decision variable:  $r(t_m) = \pm p_o(t_m) + n_o(t_m)$ , where  $n_o(t_m) \sim \mathcal{N}(0, \sigma_n^2)$
- Let  $\pm p_o(t_m) = \pm A_p$ . Then  $r_0(t) \sim \mathcal{N}(-A_p, \sigma_n^2)$  and  $r_1(t) \sim \mathcal{N}(A_p, \sigma_n^2)$
- Optimum detection threshold? Probability of bit error  $P_e$ ?

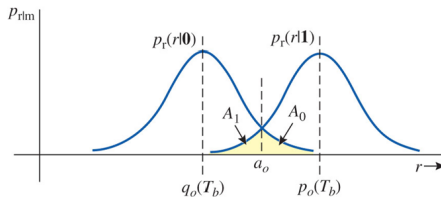
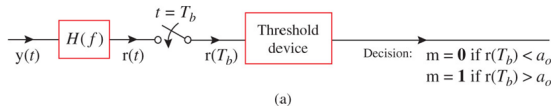


# MATCHED FILTER

- Want to minimize  $P_e$  using best possible receiver filter  $H(f)$
- Did you follow the matched filter derivation in the reading (Section 9.1.2)?
- Cauchy-Schwarz inequality!
- General matched filter:  $H(f) = \frac{P^*(f)e^{-j2\pi ft_m}}{S_n(f)}$
- For AWGN:
  - $S_n(f) = N_0/2$  where  $N_0$  (or  $\mathcal{N}$  in the textbook) is the noise PSD (or a measure of the total noise power in 1 Hz of bandwidth)
  - $H(f) = P^*(f)e^{-j2\pi ft_m}$  and  $h(t) = p(t_m - t)$  for any real valued pulse  $p(t)$
  - $t_m$  becomes  $T_0$  since  $p_0(t) = h(t) * p(t)$  reaches its max amplitude at  $t = T_0$
  - $H(f) = P^*(f)e^{-j2\pi fT_0}$  and  $h(t) = p(T_0 - t)$
  - $P_e = Q\left(\sqrt{\frac{2E_p}{N_0}}\right)$ , where  $E_p = \int_0^{T_0} |p(t)|^2 dt$  is the energy of the pulse
- Correlation detector produces equivalent result as matched filter - uses cross-correlation to measure similarity of received signal with  $p(t)$

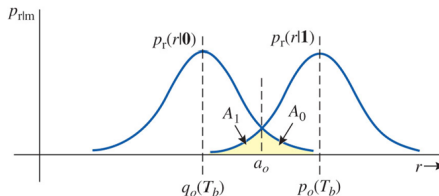
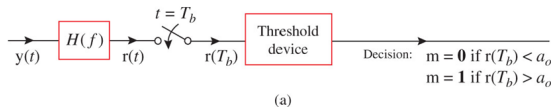
# BINARY SIGNALING

- Generalized form of what we've already covered - allow two different transmit pulses,  $p(t)$  or  $q(t)$  (representing 1 or 0)
- Both have same noise  $n(t)$  added to them
- $p_{r|m} \sim$  Gaussian with mean either  $q_0(T_b)$  or  $p_0(T_b)$  and variance  $\sigma_n^2$
- Optimal threshold:  $a_0 = \frac{p_0(T_b) + q_0(T_b)}{2}$  (assuming equally likely transmission, i.e.,  $p_m(0) = p_m(1) = 0.5$ )



# BINARY SIGNALING (CONT'D)

- Optimum filter:  $H(f) = [P^*(f) - Q^*(f)]e^{-j2\pi fT_b}$  and  $h(t) = p^*(T_b - t) - q^*(T_b - t) \rightarrow$  matched to pulse  $p(t) - q(t)$
- $P_e = P_b = Q \left( \sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}} \right)$ , where  $E_{pq} = \text{Re} \left\{ \int_0^{T_b} p(t)q^*(t)dt \right\}$
- For real values pulses,  $h(t) = p(T_b - t) - q(T_b - t)$



# BINARY SIGNALING PERFORMANCE (AWGN)

**Polar signaling,  $q(t) = -p(t)$**

